

Xaridotis Model Routing Architecture

#ai #gemini-bot #chatbot #xaridotis

Architecture Model Routing

Layer A: The Curator(Routing & Triage; Classifies intent, complexity, and memory requirements instantly on webhook hit):

Tier 1 (Primary): @cf/mistralai/mistral-small-3.1-24b-instruct (via runCfAi)

Tier 2 (Fallback): @cf/meta/llama-3.3-70b-instruct-fp8-fast (via runCfAi)

Tier 3 (Fallback): @cf/qwen/qwen2.5-coder-32b-instruct (kept as proven safety net)

Tier 4 (Fallback): claude-haiku-4-5 with structured outputs via Anthropic direct (make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family)

(Casing strict equality 'toLowerCase().trim()' in strictParse';

Crisis regex on raw user text; force-promote if Curator missed it; mapToLegacy source labels2 values4 + +crisis-backstop suffix)

Layer B1: The "Soul" (Standard Chat; Curator flags complexity: casual or intent: low and no heavy emotional background.

Handles standard casual chat, quick replies, and transactional talk):

Tier 1 (Primary): gemini-3.1-flash-lite (Google API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API)

Tier 3 (Fallback): gpt-5.4-mini (reasoning medium; via OpenAI API)

Tier 4 (Fallback): Claude Haiku 4.5 via Anthropic direct

Tier 5 (Fallback): @cf/google/gemma-4-26b-a4b-it (Workers AI; ensure the configurations are set up accordingly, as configurations different to Gemini model from the direct API, ensure using full capabilities, let the model decide the thinking level)

Layer B2: The "Soul" (Deep Emotional: Handles venting, emotional continuity, and persona-critical moments. Triggered by Curator complexity: high) &

Layer C: The Brain (Curator flags intent: {crisis, code}):

Tier 1 (Primary): gemini-pro-latest' (Google API; thinking level default to let the model decide);

Tier 2 (Fallback): gemini-3.5-flash (Google API; thinking level default to let the model decide);

Tier 3 (Fallback): gpt-5.4 (reasoning medium; via OpenAI API; make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family and Anthropic)

Tier 4 (Fallback): claude-sonnet-4-6 (via Anthropic API; make sure routed to a correct

configuration for input (schema) to get appropriate output (will be different to Gemini family);

Tier 5 (Fallback): @cf/google/gemma-4-26b-a4b-it (Workers AI; ensure the configurations are set up accordingly, as configurations different to Gemini model from the direct API, ensure using full capabilities, thinking level high);

Layer D: The Body (Curator flags intent: functional (timers, pins, checklists):

Tier 1 (Primary): gemini-3.1-flash-lite (Google API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): gpt-5.4-mini (reasoning medium; via OpenAI API)

Tier 4 (Fallback): @cf/google/gemma-4-26b-a4b-it (Workers AI; ensure the configurations are set up accordingly, as configurations different to Gemini model from the direct API, ensure using full capabilities, thinking default to let the model decide;

Layer F3: Subconscious (Factual Extraction: fires inside the chat handler after every message for the Knowledge Graph):

Tier 1 (Primary): gemini-3.1-flash-lite (Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): gemini-pro-latest (thinking level: default; Google by API);

Layer F4: Subconscious (Therapeutic REM):

Tier 1 (Primary): gemini-pro-latest (thinking level: default; Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): claude-sonnet-4-6 (via Anthropic API; make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family);

Tier 4 (Fallback): gpt-5.5 (reasoning medium default); via OpenAI API; make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family and Anthropic), check appropriate documents;

generateShortResponse (greetings, ack replies, ad-hoc system AI messages):

Tier 1 (Primary): gemini-3.1-flash-lite (Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): @cf/google/gemma-4-26b-a4b-it (Workers AI; ensure the configurations are set up accordingly, as configurations different to Gemini model from the direct API, ensure using full capabilities, thinking default to let the model decide;

background-tasks FINAL:

Curator (pre-response context analysis):

Tier 1 (Primary): gemini-3.1-flash-lite (thinking level: default; Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): gpt-5.4-mini (reasoning medium; via OpenAI API)

Conversation mode tagger (legacy fallback):

Tier 1 (Primary): gemini-3.1-flash-lite (thinking level: default; Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): gpt-5.4-mini (reasoning medium; via OpenAI API)

Mood score ack (after user taps mood score) + **Mood emotions ack** (after user picks emotions) + **Mood synthesis** (end-of-flow therapeutic observation):

Tier 1 (Primary): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 2 (Fallback): gpt-5.4 (reasoning medium; via OpenAI API; make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family and Anthropic)

Tier 3 (Fallback): gemini-3.1-flash-lite (thinking level: default; Google by API);

Triple extraction (knowledge graph) + **Mood tagging** (clinical tags) + Memory deduplication + Style card consolidation + Persona evolution:

Tier 1 (Primary): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 2 (Fallback): gpt-5.4 (reasoning medium; via OpenAI API; make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family and Anthropic)

Tier 3 (Fallback): gemini-3.1-flash-lite (thinking level: default; Google by API);

Reaction interpretation (emoji feedback):

Tier 1 (Primary): gemini-3.1-flash-lite (thinking level: default; Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): gpt-5.4-mini (reasoning medium; via OpenAI API)

Auto-episode logger + factual triples (background extract):

Tier 1 (Primary): gemini-3.1-flash-lite (thinking level: default; Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): gpt-5.4-mini (reasoning medium; via OpenAI API)

Memory consolidation workflow (nightly):

Tier 1 (Primary): gemini-pro-latest (thinking level: default; Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): claude-sonnet-4-6 (via Anthropic API; make sure routed to a correct

configuration for input (schema) to get appropriate output (will be different to Gemini family);
Tier 4 (Fallback): gpt-5.5 (reasoning medium default); via OpenAI API; make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family and Anthropic), check appropriate documents;

Vectorize embedding:

Tier 1 (Primary): gemini-embedding-2 (thinking level: default; Google by API: <https://ai.google.dev/gemini-api/docs/embeddings>)

TTS:

Tier 1 (Primary): gemini-2.5-pro-preview-tts (Google by API);

Tier 2 (Fallback): Gemini 3.1 Flash TTS (Google API);

Voice transcription:

Tier 1 (Primary): gemini-pro-latest (thinking level: default; Google by API);

Tier 2 (Fallback): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 3 (Fallback): gemini-3.1-flash-lite (thinking level: default; Google by API);

HEAVY_BG_TIERS, Chat_summaries Compression Model Tiers:

Tier 1 (Primary): gemini-3.5-flash (thinking level default, let the model decide (Google API);

Tier 2 (Fallback): gemini-pro-latest (thinking level: default; Google by API);

Tier 3 (Fallback): gpt-5.4 (reasoning medium; via OpenAI API; make sure routed to a correct configuration for input (schema) to get appropriate output (will be different to Gemini family and Anthropic)

Tier 4 (Fallback): gemini-3.1-flash-lite (thinking level: default; Google by API);

<https://ai.google.dev/gemini-api/docs/flex-inference> should be implemented later